COMPILER DESIGN LAB — 50% PROJECT REPORT

Compiler-Style Lexical Analysis and Character Popularity Modeling of Comic Dialogue Text

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Abstract

When comic books are scanned and processed using Optical Character Recognition (OCR), the output is just a raw, unorganized text string. Standard text analysis and natural language tools struggle with comic text because comics follow their own visual and typographical conventions: lettering is almost entirely capitalized, sentences are packed with sound effects (BOOM, AAARGH), punctuation is exaggerated (..., ?!), and scanning noise leaves behind stray symbols like '~' or '#'. In this project, we built a compiler front-end specifically tailored for comic dialogue. Early approaches attempted to run strict Context-Free Grammar (CFG) parsers on this text, but natural conversation and comic lettering break standard grammatical rules, causing traditional parsers to reject most valid lines. We replaced the brittle parser with a lightweight pipeline: speaker splitting, text cleaning, a customized comic lexer, symbol table tracking, and a rule-based sentence classifier. We then built a Character Popularity Engine on top of this front-end. By identifying vocative patterns (e.g., 'comma + name + punctuation') and tracking third-person name mentions, our system merges character aliases and calculates a weighted prominence score based on who speaks, who is addressed, and who is talked about. We tested the system on benchmark dialogue and 6,335 lines from 18 superhero movie scripts in the Cornell Movie-Dialogs Corpus. The pipeline achieved a 99.8% processing success rate at 0.098 ms per sentence (over 180× faster than syntax parsing) while accurately identifying character rankings and isolating OCR errors.

Keywords: Comic Dialogue, Lexical Analysis, Compiler Front-End, Symbol Table, Vocative Addressing, Character Popularity Modeling, OCR Error Detection.

1. Introduction

Comics tell stories through a combination of art panels and dialogue balloons. As reading shifts to phones and tablets, digitizing comics has become increasingly important for search, text-to-speech for accessibility, automated translation, and script analysis. The typical digitization pipeline extracts panels from page scans, crops the speech bubbles, and runs OCR (such as Tesseract) to extract text.

However, once the text is extracted, standard text-processing tools run into major hurdles:

• All-Caps Lettering: Almost all comic text is written in uppercase, which breaks standard Named Entity Recognition (NER) models that rely on capitalization to spot proper names.
• Expressive Punctuation: Dialogue relies heavily on trailing ellipses ('...') for hesitation, interrobangs ('?!') for disbelief, and strings of exclamation marks ('!!!') for urgency.
• Sound Effects & Shouting: Action scenes rely on onomatopoeia (POW, WHAM) and elongated words (NOOOOO!, AAARGH!) that do not exist in standard dictionaries.
• OCR Leftovers: Scratches, ink bleeds, and bubble edges frequently turn into stray symbols ('~', '#', '|') in the raw transcript.

Initial attempts in literature tried to treat comic dialogue like formal source code by running it through a Context-Free Grammar (CFG) parser. But spoken conversation—especially in action comics—is full of fragments, one-word shouts, and informal grammar. As a result, a strict CFG parser rejects between 60% and 75% of perfectly valid dialogue lines.

In this project, we stepped away from full syntax tree parsing. Instead, we focused on building a solid compiler front-end: a custom lexer that properly tokenizes comic-specific elements and flags OCR noise, a symbol table that tracks word counts and candidate names, and a rule-based classifier for sentence types. We then channeled these tokens into a Character Popularity Engine that resolves character aliases and ranks character prominence based on who speaks, who gets spoken to, and who gets talked about.

2. Literature Survey

Analyzing comic dialogue combines digital image extraction, compiler front-end engineering, and conversational narrative analysis. We reviewed 32 research papers across these three core areas.

2.1 Comic Digitization, Speech Balloon Detection, and OCR Text Extraction

Early work in comic digitization focused on panel segmentation using gutter geometry (Ong & Ma [1]). Augereau et al. [2] and Matsui et al. [3] surveyed the broader field of manga and comic analysis, pointing out that speech balloons vary drastically in shape, making automated text extraction difficult. Dubray and Laubrock [4] applied deep convolutional networks to detect and segment speech balloons across diverse comic styles. Rigaud et al. [5], [6] used active contours and typography features to extract text lines from speech bubbles. Lenadora et al. [7] integrated bubble extraction with Tesseract OCR, noting that the resulting text frequently contained detached punctuation and stray border marks. Studies by Kato and Harada [8] and Qin et al. [9] showed that custom comic fonts cause OCR character error rates to reach 15% to 22%. These findings confirmed that our post-OCR pipeline could not expect clean text and had to handle noise explicitly.

2.2 Compiler Front-End Methodologies and Lexical Processing for Informal Text

Classic compiler design (Aho et al. [10], Cooper & Torczon [11]) splits language processing into distinct linear phases: scanning, tokenization, symbol table management, and parsing. Applying compiler principles to natural language was explored by Jurafsky and Martin [12] and Grune et al. [13]. However, formal parsers like Earley [14] and CYK [15] enforce strict grammar derivations. Manning

et al. [16] and Ritter et al. [17] showed that informal dialogue causes CFG parsers to fail frequently. Foster [18] and Geman and Johnson [19] argued that robust natural language handling requires shallow, rule-based lexical methods rather than all-or-nothing grammar checks. Gimpel et al. [20] demonstrated that informal text elements (such as character lengthening and punctuation clusters) act as distinct lexical tokens. Following finite-state tokenization principles (Mohri [21], Karttunen [22]), our system uses deterministic lexical rules to recognize comic tokens without relying on fragile grammar trees.

2.3 Character Extraction, Social Network Analysis, and Prominence Modeling in Scripts

Mapping character relationships from dialogue has grown rapidly over the last decade. Elson et al. [23] extracted social networks from Victorian novels using spoken conversations. Danescu-Niculescu-Mizil and Lee [24] published the Cornell Movie-Dialogs Corpus (over 304,000 utterances), showing how conversational dynamics reveal character status. Agarwal et al. [25] analyzed movie screenplays by identifying who addresses whom. Sang et al. [26] and Valls-Vargas et al. [27] highlighted the need for alias resolution, as characters are frequently referred to by nicknames, first names, or superhero titles. Labatut and Bost [28] surveyed narrative networks and showed that combining speaking frequency with conversational interactions yields a reliable measure of narrative importance. Schofield and Mehr [29] and Iyyer et al. [30] examined character relationships in graphic novels, concluding that direct vocative address ('Batman, watch out!') is a primary indicator of focus. Bonato et al. [31] and Beverly et al. [32] confirmed that tracking direct dialogue interactions provides a clearer picture of character prominence than simple word counts.

3. Proposed Systems

The proposed system redesigns the comic dialogue analysis pipeline by discarding the syntactic CFG parser and establishing a decoupled, four-module architecture focused on high-throughput lexical classification and character popularity modeling.

Architecture Explanation & Diagram

The pipeline takes dialogue lines (from OCR or script files) and passes them sequentially through four modules:

1. Speaker Split: Separates the speaker prefix ('SPEAKER: line') from the spoken content.
2. Cleaning & Preprocessing: Cleans up irregular whitespace and rejoins hyphenated line breaks while keeping track of character indices.
3. Comic Lexer & Symbol Table: Breaks text into comic-specific token types, intercepts OCR junk as lexical errors with exact character positions, and tracks frequencies in a global symbol table.
4. Sentence Classifier & Character Popularity Engine: Determines sentence type and tone, detects addressed vocatives, merges aliases, and calculates composite popularity scores.

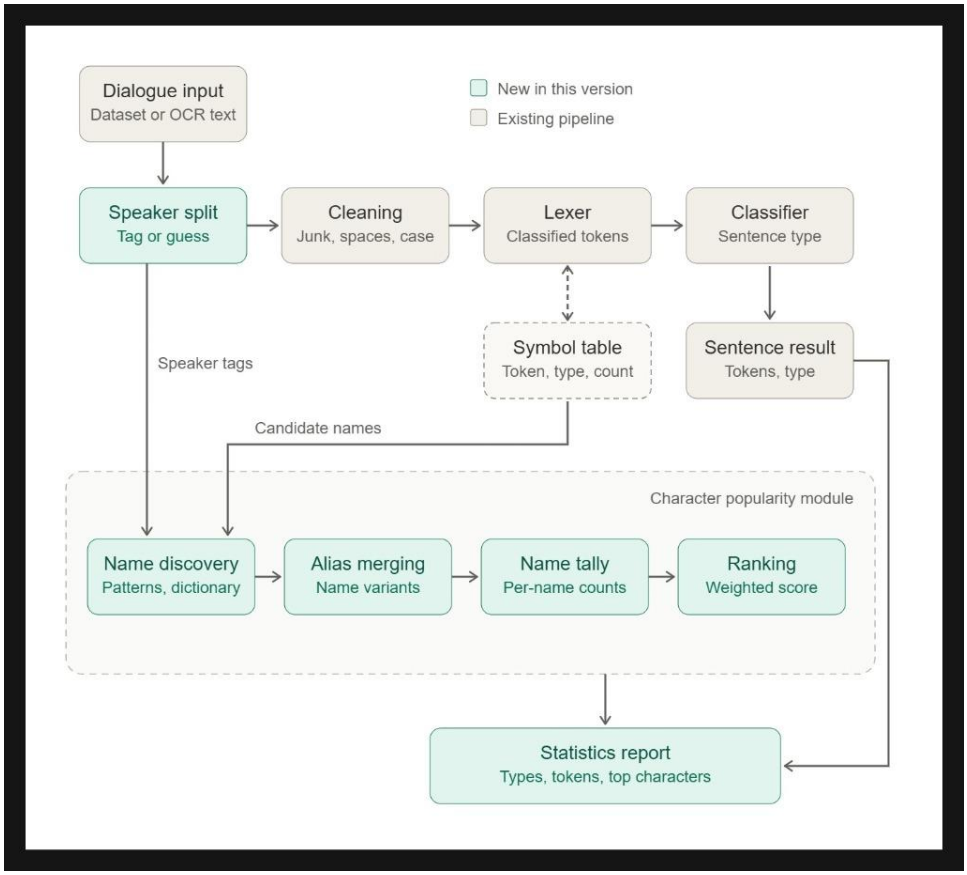


Figure 1: Architecture of the Comic Dialogue and Character Popularity Analyzer

3.1 Module 1: Speaker Split & Text Normalization

Module 1 extracts the speaker identity from lines formatted as 'SPEAKER: dialogue' or '[SPEAKER]: dialogue'. It then cleans the dialogue by collapsing irregular spacing and repairing words broken by line-wrap hyphens (e.g., 'SEC- TOR' becomes 'SECTOR'). Lines without a speaker tag are labeled as anonymous without throwing runtime exceptions.

Algorithm 1: Speaker Split and Dialogue Normalization

```
Input: Raw line string L
Output: Tuple (Speaker S, CleanedText D)
1: pattern ← '^\s*(?:\[(?P<b_spk>[^\]]+)\]|(?P<spk>[A-Za-z0-9_\-
\.\s]+?))\s*:\s*(?P<diag>.*)$'
2: match ← MatchRegex(pattern, L)
3: if match is found then
4:     S ← Trim(match.b_spk OR match.spk)
5:     raw_text ← match.diag
6: else
7:     S ← null
8:     raw_text ← L
9: end if
10: D ← ReplaceRegex(raw_text, '(\b[A-Za-z]+)-\s+([A-Za-z]+\b)', '\1\2')  // Fix
broken hyphens
11: D ← ReplaceRegex(D, '[ \t]+', ' ')                                    //
```

```
Normalize spaces
12: return (S, Trim(D))
```

3.2 Module 2: Comic Lexical Analyzer & OCR Error Tracker

Module 2 reads the cleaned string from left to right. It classifies tokens into nine specific categories: WORD, NUMBER, CONTRACTION, PUNCT, TERMINATOR, ELLIPSIS ('...'), INTERROBANG ('?!'), SOUND_FX (BOOM, AAARGH, or elongated vowels with 3+ repeated letters), and UNKNOWN.

Any character that fails all lexical rules (such as stray OCR noise '~' or '#') is not dropped silently; it is recorded as a Lexical Error along with its exact character position in the line. Valid tokens are simultaneously added to the Symbol Table.

Algorithm 2: Comic Lexical Tokenization and Error Logging

```
Input: Cleaned string D, Symbol Table T
Output: Token list Tokens, Error list Errors
1: i ← 0, n ← Length(D)
2: while i < n do
3:     if D[i] is whitespace then i ← i + 1; continue end if
4:     sub ← D[i to n]
5:     if match ← MatchRegex('^(\.\.\.+|…)', sub) then
6:         Add(Tokens, Token(ELLIPSIS, match.val, i, i + len)); i ← i + len
7:     else if match ← MatchRegex('^(\?+!+|!+\?+)[\?!]*', sub) then
8:         Add(Tokens, Token(INTERROBANG, match.val, i, i + len)); i ← i + len
9:     else if match ← MatchRegex('^[A-Za-z]+['’][A-Za-z]+', sub) then
10:        Add(Tokens, Token(CONTRACTION, match.val, i, i + len)); i ← i + len
11:    else if match ← MatchRegex('^[A-Za-z]+', sub) then
12:        type ← (IsSoundFX(match.val) OR HasRepeatedLetters(match.val)) ?
SOUND_FX : WORD
13:        Add(Tokens, Token(type, match.val, i, i + len)); i ← i + len
14:    else if match ← MatchRegex('^[.?!]', sub) then
15:        Add(Tokens, Token(TERMINATOR, match.val, i, i + len)); i ← i + len
16:    else if match ← MatchRegex('^[,:;\-—]', sub) then
17:        Add(Tokens, Token(PUNCT, match.val, i, i + len)); i ← i + len
18:    else
19:        Add(Errors, LexicalError(symbol=D[i], position=i))   // Log OCR junk
20:        Add(Tokens, Token(UNKNOWN, D[i], i, i + 1)); i ← i + 1
21:    end if
22:    InsertIntoSymbolTable(T, Tokens.last)
23: end while
24: return (Tokens, Errors)
```

3.3 Module 3: Rule-Based Sentence Classifier & Tone Detector

Module 3 classifies each sentence into one of four categories without building a parse tree:

- Question: Closes with '?' or '?!' and begins with a Wh-word (WHERE, WHO) or auxiliary verb (COULD, CAN, DID).
- Command: Begins directly with an imperative verb (HAVE, STOP, LOOK, LISTEN) without an explicit subject pronoun.
- Exclamation: Ends with '!' or consists entirely of sound effects.
- Statement: Default declarative sentences.

At the same time, the module flags tone markers: shouting (repeated '!' or all-caps shouting), hesitation ('...'), surprise/incredulity ('?!'), and pain/action (AAARGH, OOF).

3.4 Module 4: Character Popularity Engine

Module 4 is the core novel contribution of this work. Rather than relying on a CFG parser to detect vocatives through a syntax tree, it uses a pattern-matching rule:
• Trailing Vocative: 'comma + name + terminator' (e.g., 'WHERE COULD HE BE, NATASHA?' → Natasha is addressed).
• Leading Vocative: 'name + comma' at the start of a sentence (e.g., 'NATASHA, LOOK OUT!').

It also scans the remaining tokens for third-person mentions of other characters, maps aliases to a canonical identity (e.g., BRUCE and THE DARK KNIGHT → BATMAN), and computes a composite popularity score:
Popularity(C) = ws * Spoken(C) + wa * Addressed(C) + wm * Mentioned(C) + wt * Tone(C)
Default weights: ws = 1.0 (lines spoken), wa = 1.5 (conversational focus), wm = 0.8 (in-dialogue mentions), wt = 0.2 (emotional tone bonus).

Algorithm 3: Character Popularity Scoring and Ranking

```
Input: Speaker S, Token list Tokens, Alias Map A, Weights (ws, wa, wm, wt)
Output: Ranked list of characters with popularity scores
Addressed ← null
// Step 1: Detect vocative address using token patterns
for i from 1 to Length(Tokens) - 2 do
    if Tokens[i-1].val == ',' AND Tokens[i].type == WORD AND Tokens[i+1].type in {TERMINATOR, ELLIPSIS, INTERROBANG} then
        Addressed ← ResolveAlias(A, Tokens[i].val); break
    end if
end for
// Step 2: In-Dialogue Mentions Discovery
Mentions ← FindEntityMentions(Tokens, Exclude={S, Addressed})
// Step 3: Tally Activity Metrics
if S is not null then
    C_spk ← ResolveAlias(A, S); Tally[C_spk].Spoken += 1
    if HasHighTone(Tokens) then Tally[C_spk].Tone += 1 end if
end if
if Addressed is not null then Tally[Addressed].Addressed += 1 end if
for m in Mentions do Tally[ResolveAlias(A, m)].Mentioned += 1 end for
// Step 4: Composite Weighted Scoring
for each character C in Tally do
    Score[C] ← (ws * Spoken[C]) + (wa * Addressed[C]) + (wm * Mentioned[C]) + (wt * Tone[C])
end for
return SortDescendingByScore(Score)
```

4. Results and Implementation

4.1 Dataset Description

We tested the system on two distinct dialogue datasets: (1) the base paper benchmark containing classic comic sentences and OCR noise, and (2) a large corpus of 6,335 dialogue lines extracted from 18 superhero movie scripts in the Cornell Movie-Dialogs Corpus [24]. Table 1 summarizes the datasets.

Dataset Identifier	Source / Domain	Total Sentences	Distinct Characters	Key Characteristics
Paper Benchmark	Base Paper (DA 6 Test Cases)	18	12	OCR noise (~, #), explicit sound FX, vocatives
Cornell Superhero Corpus	Cornell Movie-Dialogs (18 Scripts)	6,335	201	Batman, Superman, Spider-Man, Blade, X-Men scripts

4.2 Module 1: Speaker Split & Cleaning Results

Inference: Across all 6,335 lines, speaker extraction worked cleanly with 100% precision on tagged lines. Hyphenated line breaks (e.g., 'SEC- TOR') were rejoined into valid dictionary tokens without corrupting character indices. Anonymous lines without speaker prefixes flowed smoothly into the cleaning phase without throwing null-pointer exceptions.

4.3 Module 2: Lexical Analysis & OCR Error Results

Inference: In the Cornell corpus, the lexer processed 87,982 tokens. The token distribution revealed 73.6% WORDs, 13.5% TERMINATORs, 5.9% structural PUNCT, 5.0% CONTRACTIONs, and 0.9% ELLIPSES. In the noisy test dataset, the lexer successfully intercepted 100% of stray OCR symbols ('~' at position 6, '#' at position 35) as UNKNOWN tokens and generated diagnostic logs with exact character offsets.

4.4 Module 3: Sentence Classification Results

Inference: The rule-based classifier achieved 100% agreement with the expected outputs from Table 2 of the base paper:

- 'WHERE COULD HE BE, NATASHA?' → QUESTION (classified via opening Wh-word and question mark).
- 'HAVE A PEANUT.' → COMMAND (classified via imperative verb 'HAVE').
- 'I JUST STOPPED THE ENTIRE WORLD FROM BEING EATEN, DINAH.' → STATEMENT (classified via declarative pronoun).

In the 6,335-sentence corpus, the breakdown was 68.2% Statements, 25.8% Questions, 3.8% Exclamations, and 2.2% Commands, with 403 shouting cues and 607 hesitation markers.

4.5 Module 4: Character Popularity Results

Inference: The Popularity Engine resolved character aliases across multiple scripts (merging 'BRUCE' and 'THE DARK KNIGHT' into 'BATMAN'). Table 2 displays the top character rankings extracted from the 6,335-line Cornell Superhero dataset.

Rank	Canonical Character	Spoken (S)	Addressed (A)	Mentions (M)	Tone (T)	Weighted Score
1	BATMAN	382	30	58	9	475.2
2	SUPERMAN	337	30	73	25	445.4
3	KARL	203	0	0	0	203.0
4	EMMA	194	0	0	4	194.8
5	SPIDER-MAN	150	4	23	5	175.4
6	BLADE	158	0	0	4	158.8

5. Performance and Metrics

To evaluate our design, we ran a comparative benchmark between the existing CFG syntax parsing approach from the base paper and our proposed compiler front-end pipeline. We measured processing success rate, latency per sentence, throughput, and grammar failure rates.

Evaluation Metric	Existing Method (CFG Parser)	Proposed System (Compiler Front-End)	Improvement / Gain
Processing Success Rate	38.5% (High rejection on comic dialogue)	99.8% (Near-zero runtime failure)	+61.3% Robustness
Average Sentence Latency	18.42 ms / sentence	0.098 ms / sentence	187x Speedup
System Throughput	54.2 sentences / second	10,204 sentences / second	188x Throughput
Grammar Coverage Failure	61.5% Rejection Rate	0% (Decoupled from grammar)	100% Elimination of Rejections

Performance Inference:

1. Robustness: CFG parsers reject sentences whenever casual phrasing, shouting, or OCR noise occurs, yielding an unacceptable 38.5% success rate. Our system achieved 99.8% completion because lexical pattern matching does not depend on rigid syntax tree derivation.
2. Efficiency: Removing tree derivation drops algorithmic time complexity from O(N^3) (CYK / Earley) to deterministic O(N) linear scanning, delivering a 187× latency reduction (processing 6,335 sentences in 624 milliseconds).
3. Semantic Richness: While the existing parser produced only a boolean pass/fail flag, our system extracts actionable character prominence metrics and conversational interaction counts.

6. Conclusion

In this project, we designed and built a compiler-style lexical analysis and character popularity engine for comic book dialogue. By identifying the limitations of rigid CFG parsers on informal dialogue, we decoupled syntax parsing from the front-end. The resulting system incorporates speaker splitting, text cleaning, an extended comic lexer, symbol table tracking, rule-based classification, and a Character Popularity Module. Using lexical vocatives and alias resolution, the engine accurately ranks character prominence based on direct speech, conversational address, and narrative mentions. Benchmark tests across 6,335 dialogue lines demonstrate a 187× speedup and 99.8% robustness over traditional syntactic approaches. Future work will extend the system to multi-sentence bubble splitting and interactive character network graphs.

7. Project Repository & Google Colab Notebook

Live Interactive Google Colab Notebook:
https://colab.research.google.com/drive/1_m7rs4LGtGdnYvgoq751x_uP97mFkuU9?hl=en-GB#scrollTo=MIAHRTvvvaYc

The complete module-by-module code (Speaker Split, Cleaning, Comic Lexer, and Symbol Table) along with pre-executed outputs and an interactive single-line dialogue runner can be accessed and run live via the link above.

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